

Lab3- Distillation

TA : Ying-Huan Chen

Due : 10/30 23:59

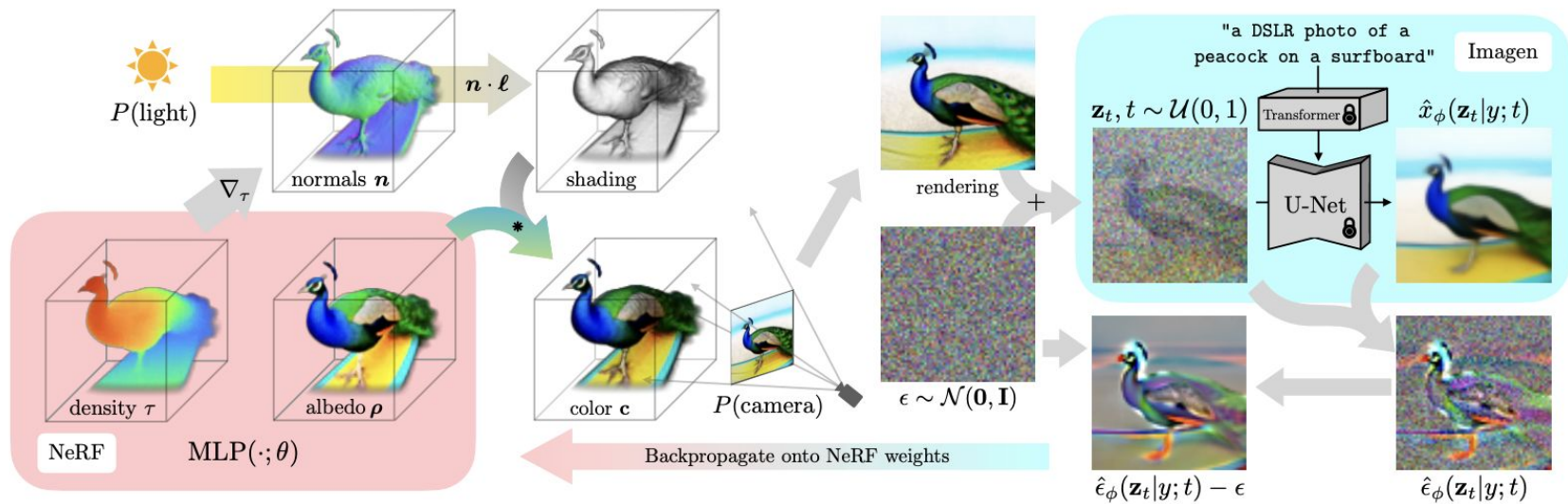
Outline

- Introduction
- TODO
- Evaluation and Grading
- Report

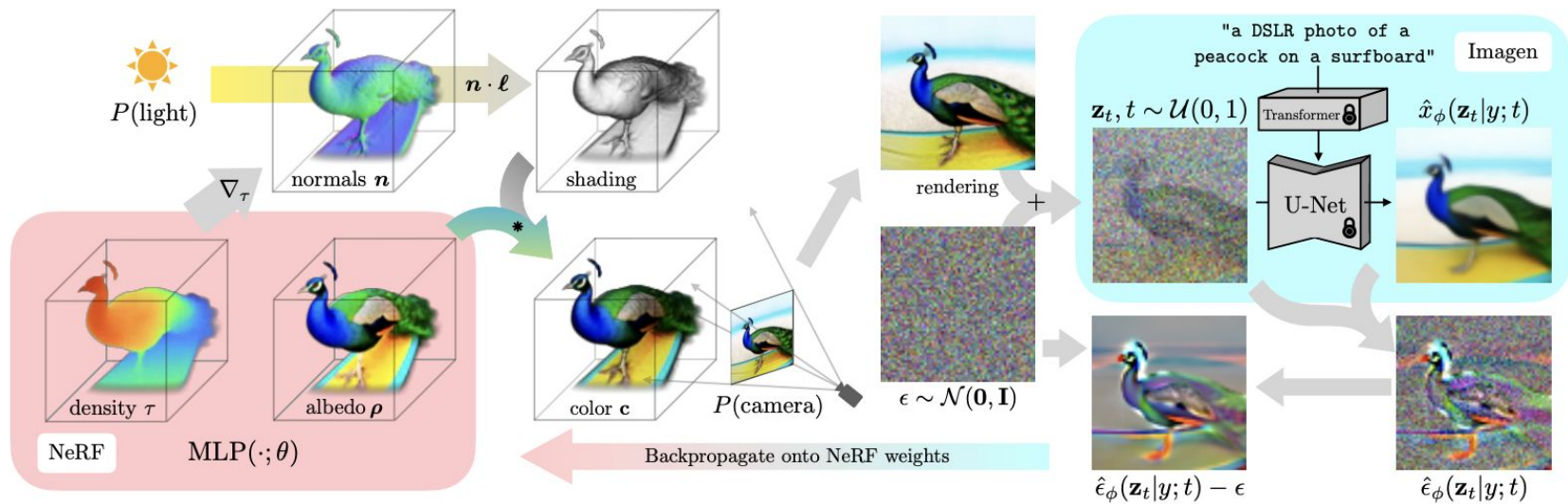
Lab3

- SDS – Score Distillation Sampling (Task 1)
- VSD – Variational Score Distillation (Task 2)
- SDI – Score Distillation via Inversion (Task 3)

SDS

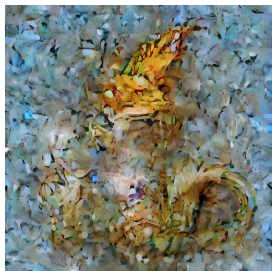


SDS



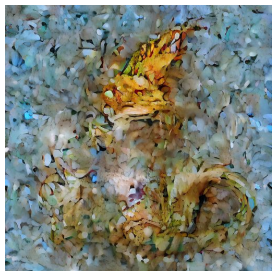
Can we directly do 2D image generation with SDS?

2D SDS



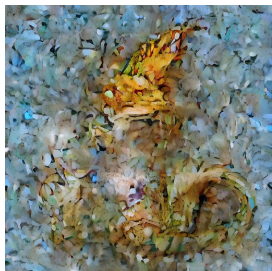
rendering latent

2D SDS



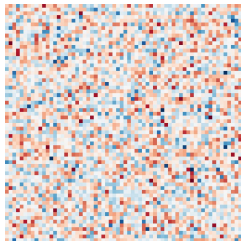
rendering latent

2D SDS



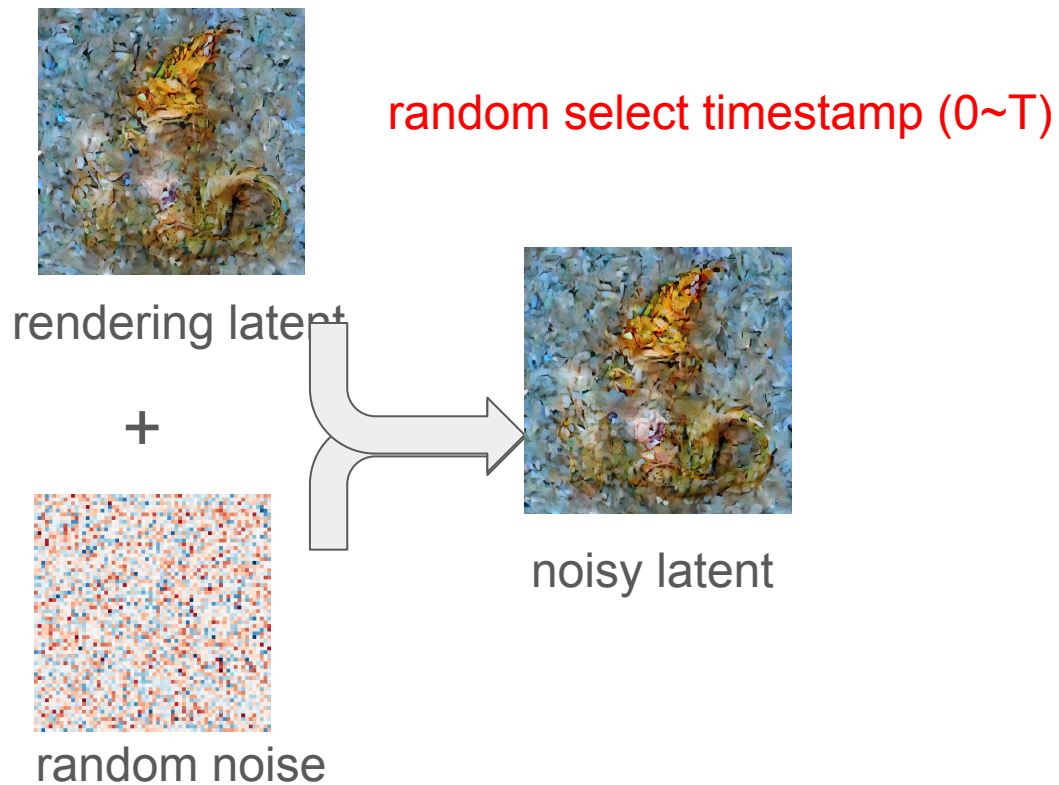
rendering latent

+

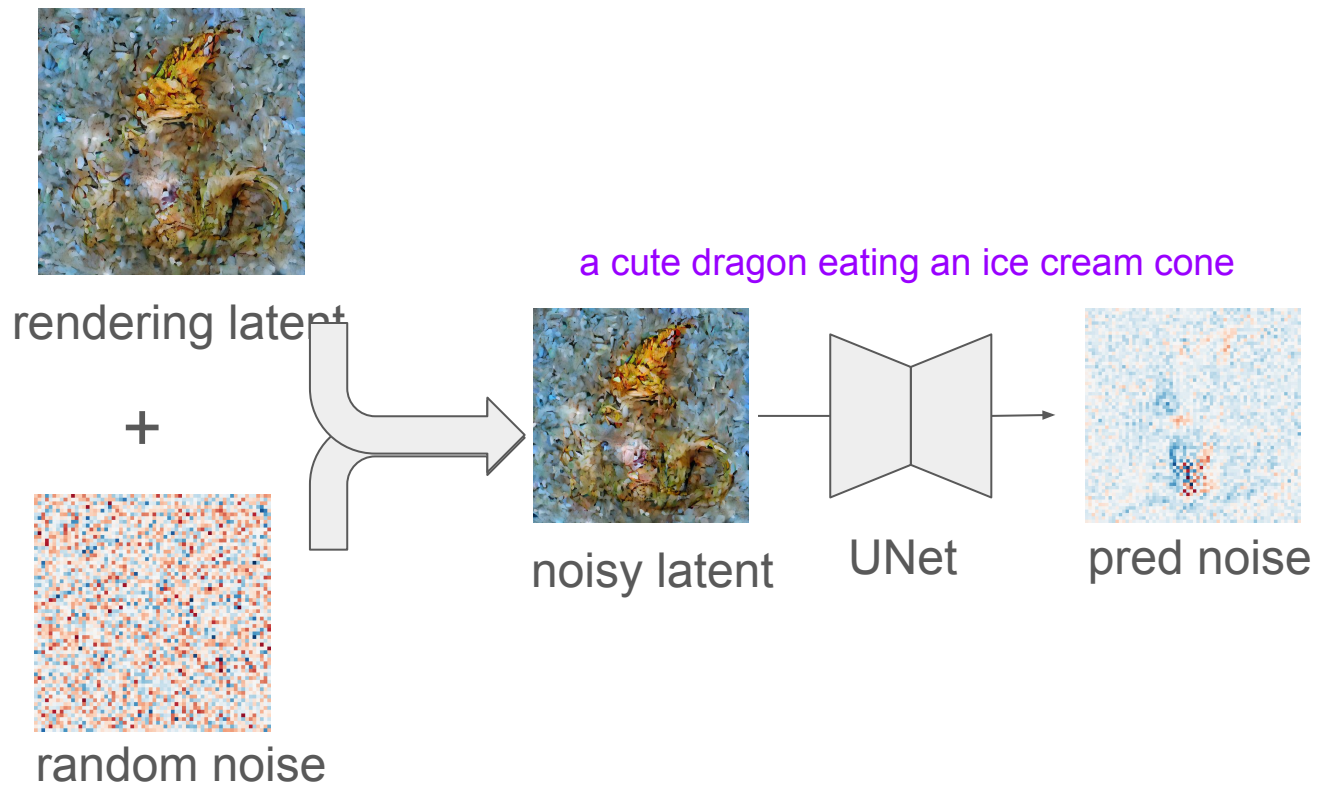


random noise

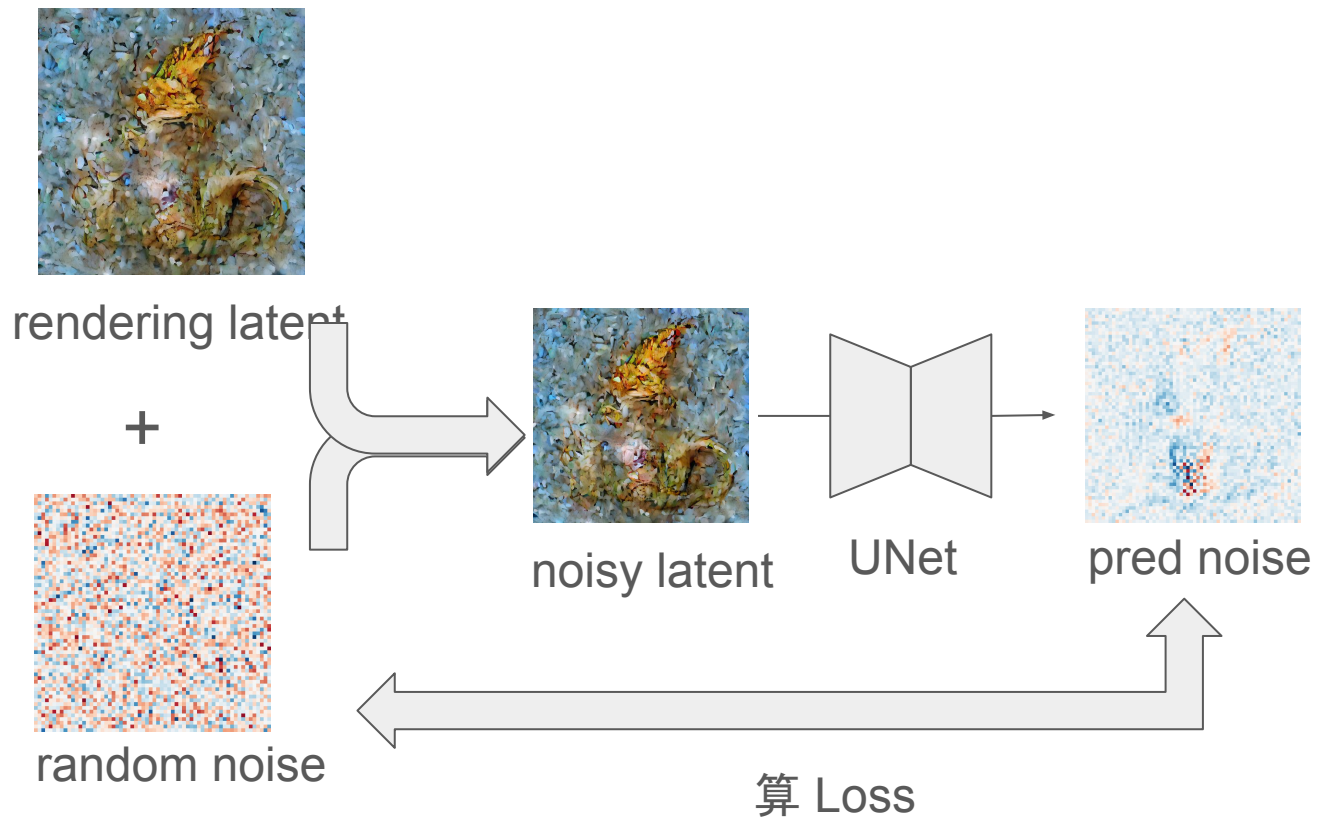
2D SDS



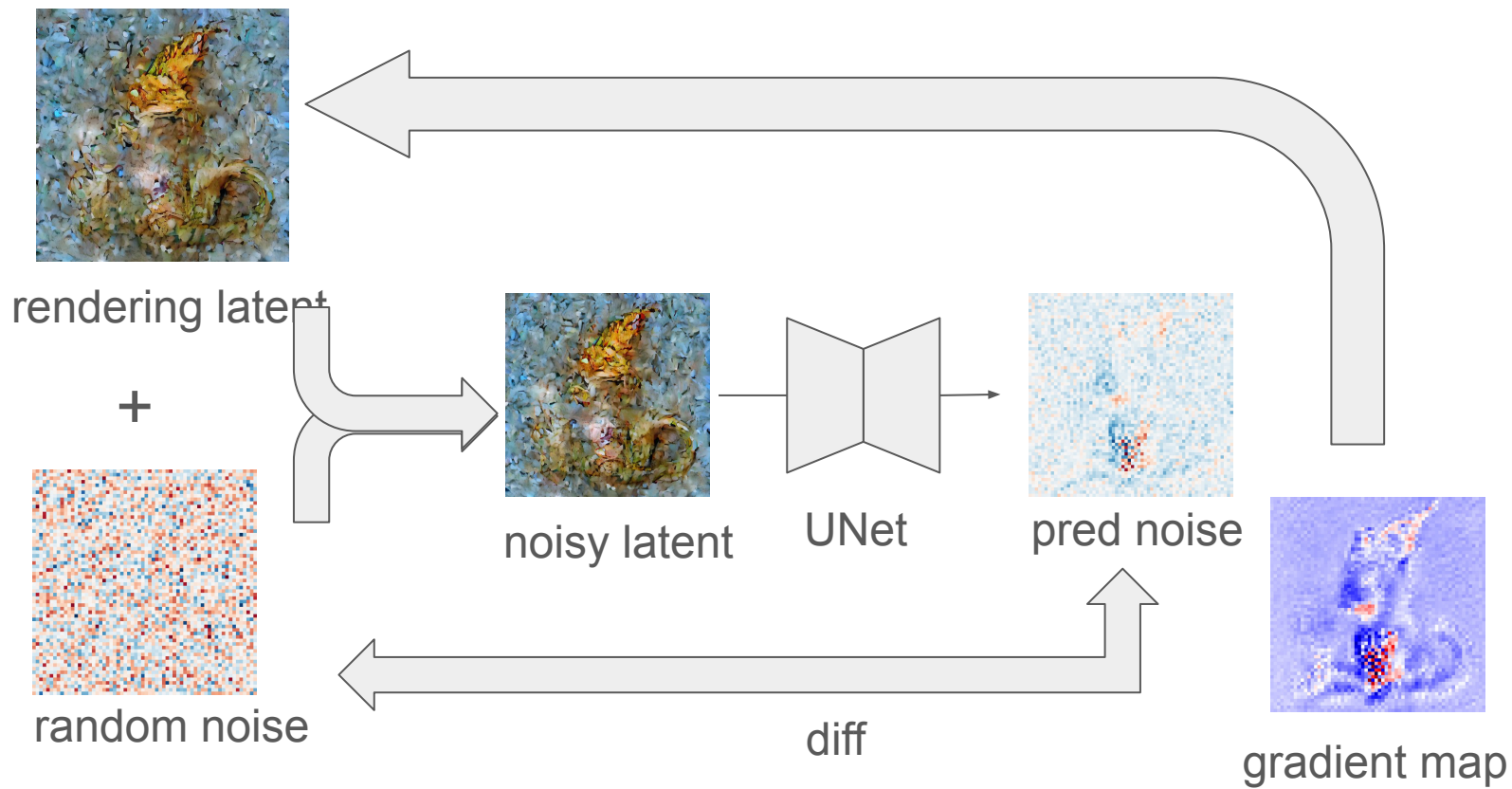
2D SDS



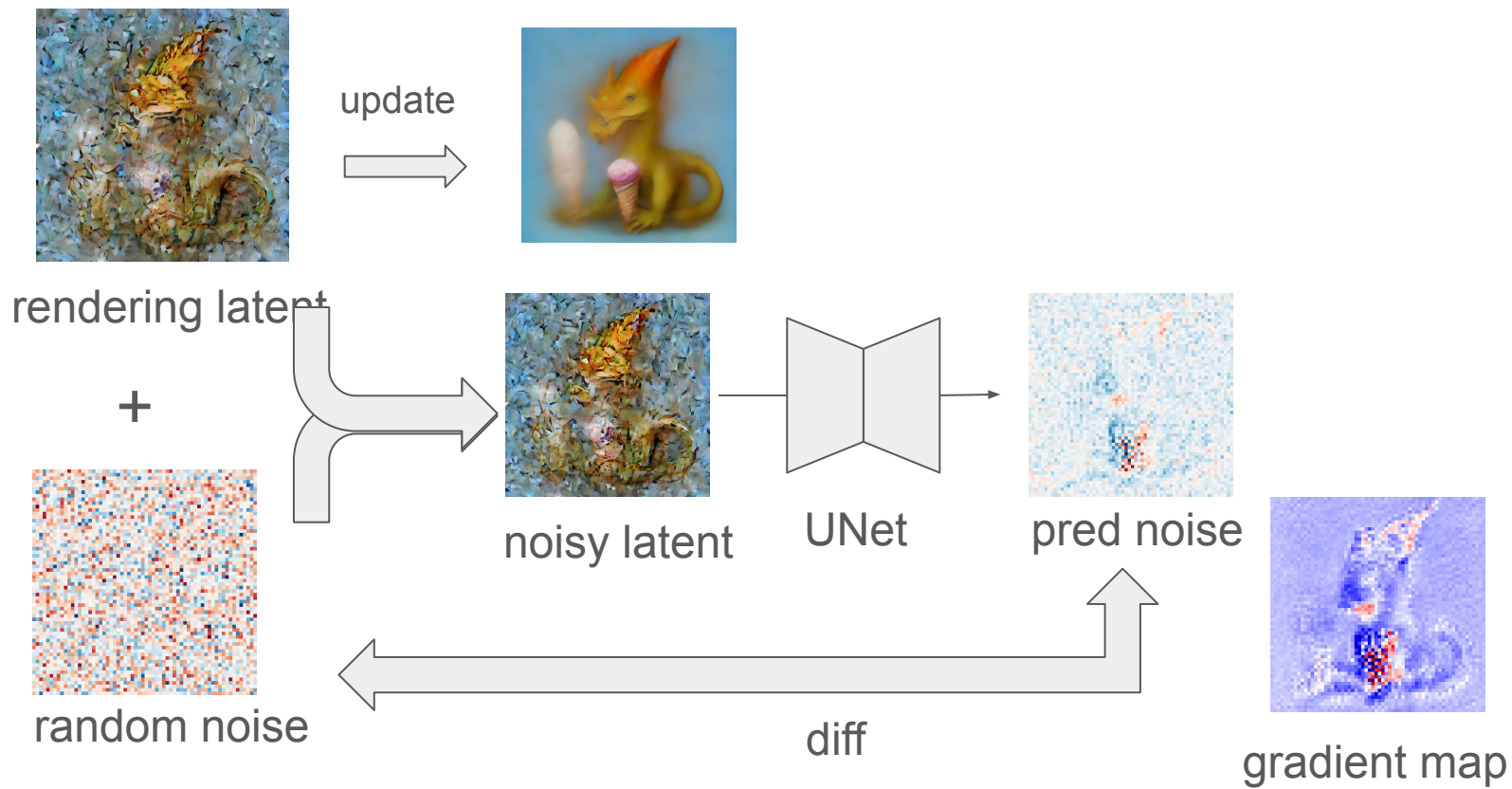
2D SDS



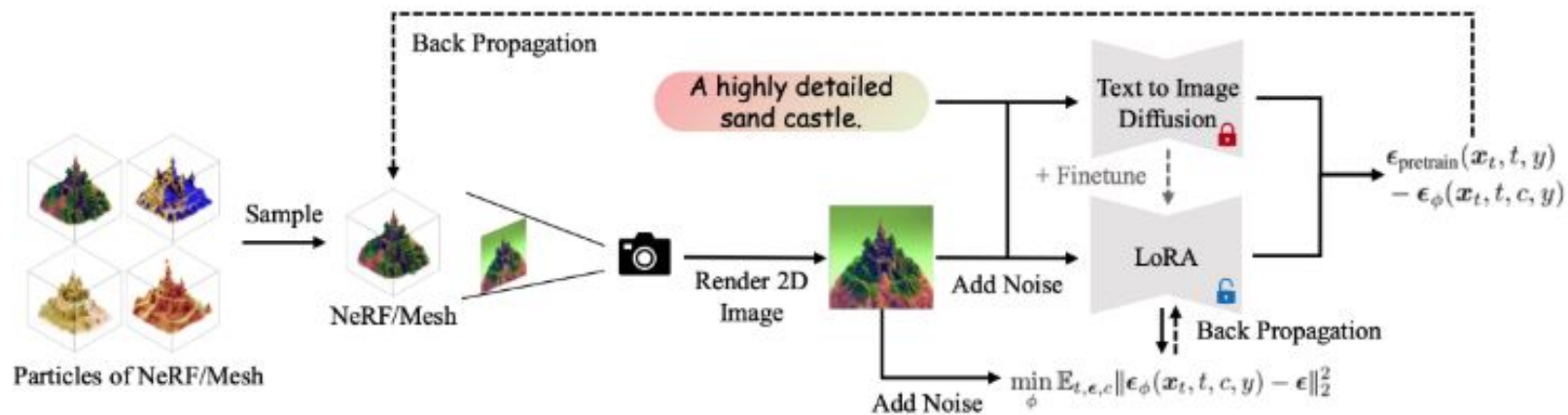
2D SDS



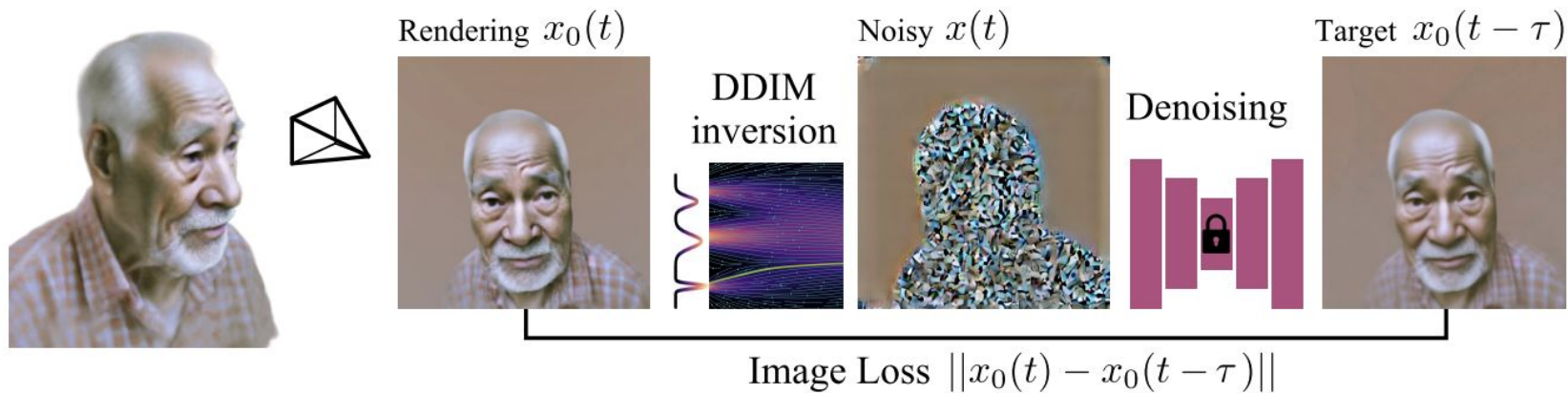
2D SDS



VSD



SDI



DDIM Inversion

- For **deterministic** sampling, the mapping from x_T to x_0 is **fixed**.
- Then, how can we also compute the **inverse mapping** from x_0 to x_T ?

給定一張照片 (x_0)，找到一個 noise (x_T)
經過 deterministic DDIM 反向去噪最終會得到原本的 x_0



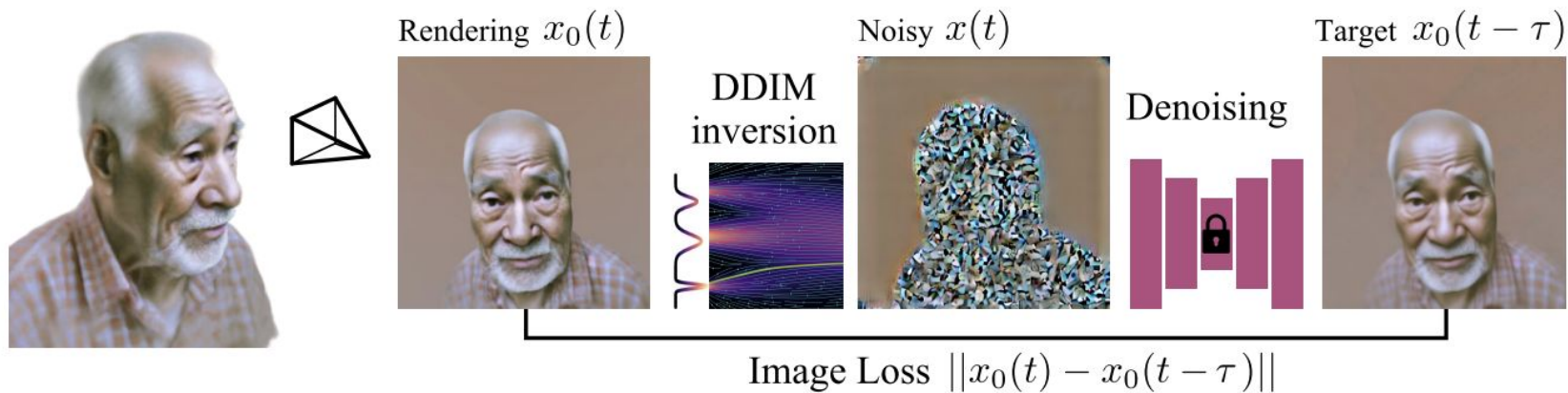
DDIM Inversion



Input Image

$$z_{t+1} = \sqrt{\frac{\alpha_{t+1}}{\alpha_t}} z_t + \left(\sqrt{\frac{1}{\alpha_{t+1}} - 1} - \sqrt{\frac{1}{\alpha_t} - 1} \right) \cdot \epsilon_\theta(z_t, t, \mathcal{C}).$$

SDI



SDI

Score Distillation via Inversion (SDI)

Additionally, decrease the time step t like the reverse process.

For each time step $t = T, \dots, 1$, repeat:

1. $c \sim \mathcal{U}(\mathcal{C})$.
 2. $\mathbf{x}_{0|t} = g(\boldsymbol{\theta}; c)$.
 3. $\mathbf{x}_t = \text{DDIM_Inversion}(\mathbf{x}_{0|t}, t)$.
 4. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}}\hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$.
 5. $\mathbf{x}_{0|t-1} = \frac{1}{\sqrt{\bar{\alpha}_{t-1}}}(\mathbf{x}_{t-1} - \sqrt{1 - \bar{\alpha}_{t-1}}\hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}, t))$.
 6. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \|\mathbf{x}_{0|t-1} - g(\boldsymbol{\theta}; c)\|^2$.
-

SDI

Score Distillation via Inversion (SDI)

Additionally, decrease the time step t like the reverse process.

For each time step $t = T, \dots, 1$, repeat:

linear decay

1. $c \sim \mathcal{U}(\mathcal{C})$.
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 4. $\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_{0|t} + \sqrt{1 - \bar{\alpha}_{t-1}}\hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$.
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-

DDIM Inversion



Input Image

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Score Distillation via Inversion (SDI)

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6. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \|\mathbf{x}_{0|t-1} - g(\boldsymbol{\theta}; c)\|^2$.

太慢了

SDI

Score Distillation via Inversion (SDI)

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6. $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \frac{\partial}{\partial \boldsymbol{\theta}} \|\mathbf{x}_{0|t-1} - g(\boldsymbol{\theta}; c)\|^2$.

實務上：每 n 個 steps 才
update 一次 target

Todo

- Ctrl-F **TODO** on guidance/sd.py

Grade

Component	Points	Description
Task 1 – SDS	20	Correct implementation + results
Task 2 – VSD	30	Correct implementation + results
Task 3 – SDI	30	Correct implementation + results
Report	20	Clear analysis, comparisons, insights
Total	100	

Evaluation

Use the unified evaluation script `eval.sh` :

```
bash eval.sh --sds
bash eval.sh --vsd --guidance 7.5 --lora-lr 1e-4 --lora-loss-weight 1.0 --lora-rank 4
bash eval.sh --sdi --guidance 7.5 --lr 0.005 --steps 1000 \
    --inversion-n-steps 10 --inversion-guidance-scale -7.5 \
    --inversion-eta 0.3 --sdi-update-interval 25
```



Each command will:

1. Generate results for all default prompts.
2. Run CLIP evaluation automatically and save final results to:

```
./outputs/{loss_type}/
├─ <prompt>.png
└─ eval.json
```



CLIP Score

CLIP Score	SDS (20 pts)	VSD/SDI (30 pts)
≥ 0.28	Full credit	Full credit
0.26–0.28	15 pts	25 pts
0.24–0.26	10 pts	20 pts
< 0.24	0	0

Report

- Specify all hyperparameters you used (per task).
- Explain SDS, VSD, and SDI — both the concept and your implementation (code).
- Compare visual and CLIP results.
- Intuitively analyze why SDI and VSD perform better than SDS (without heavy math).
- Discuss the influence of hyperparameters
 - for SDS : guidance scale.
 - for SDI : inversion steps and update interval(run ablations, and justify your findings).

Submission

Submit `{STUDENT_ID}_lab3.zip` containing:

```
./submission/  
├─ guidance/sd.py  
├─ outputs/{sds,vsd,sdi}/ + eval.json  
└─ report.pdf
```



Submission

Submit `{STUDENT_ID}_lab3.zip` containing:

```
./submission/  
├── guidance/sd.py  
├── outputs/{sds,vsd,sdi}/ + eval.json  
└── report.pdf
```



For any questions, please directly create a github issue on <https://github.com/YingHuan-Chen/Lab3-Distillation>

Reference

1. DreamFusion: Text-to-3D using 2D Diffusion — <https://arxiv.org/abs/2209.14988>
2. ProlificDreamer: Variational Score Distillation — <https://arxiv.org/abs/2305.16213>
3. Score Distillation via Inversion (SDI) — <https://arxiv.org/abs/2312.02164>
4. <https://huggingface.co/learn/diffusion-course/unit4/2>
5. <https://lukoianov.com/sdi>
6. <https://github.com/threestudio-project/threestudio>

Q&A